FUNDAMENTALS OF INSTRUCTION

FULFILLS AI.I.A/AI.I.B/AI.I.C/AI.I.D/AI.I.E/AI.I.F

Objective	
Instructor Actions	**Student Actions**
Case Studies	**Equipment**
Completion Standards	

14 NOVEMBER 2024

ANDREW TRAFICANTE

ELEMENTS

A. Human Behavior and Communication on the Learning Process
B. The Learning Process
C. The Teaching Process
D. Student Evaluation, Assessment, and Testing
E. Elements of Effective Teaching
F. Risk Management

RESOURCES

FAA-S-ACS-25 (CFI ACS) - Area I
FAA-H-8083-9B Aviation Instructor's Handbook

A. HUMAN BEHAVIOR AND COMMUNICATION ON THE LEARNING PROCESS

Elements of Human Behavior

Definitions of Human Behavior	Answers why do people function the way they do An attempt to satisfy certain needs
Instructor and Learner Relationship	Our fundamental goal as an instructor is to identify in each of our students their **motivators**, **needs**, and **defense mechanisms**, and tailor our teaching process to their learning styles.
Motivation	If I don't know what drives someone, how will I help them achieve their goals? Sure we can utilize the motivators outline in the AIH, but simply getting to know our student would immensely increase our ability to tailor the teaching process. Who wants to fill out NavLogs… Motivations lie at heart of goals
Human Needs	All of this is useless if our student isn't in the right mindset to learn. People in psychology have determined in what order our needs need to be met. **Maslow's Hierarchy** outlines when this is, and I must understand how this relates to aviation. **Self-Actualization** Need to achieve their own potential. Where is this leading to. I have an obligation to help my students. **Self-Esteem** Internal esteem - How I think about myself (instructors should promote that internal ego) External esteem - How I think others perceive me (embarrassing story) **Love and Belonging** It can be intimidating for a student to start at a new school **Safety and Security** *Does your performance degrade under extreme stress? If you felt unsafe in the cockpit, how are you supposed to learn? Spin training… Duct tape on plane on first flight… "Oh thats normal"…* **Physiological** Air, water, food, shelter, sleep *Have you ever been in class or in a meeting and really had to use the bathroom? Were you able to concentrate?*
Defense Mechanisms	Other barriers to learning also exist besides needing to meet needs. Defense mechanisms are ego protectors to unpleasant situations. **DRDRFCPR** **Denial** Refusal to accept reality. Minimization. **Repression** Bury memories inaccessible in subconscious. Irrational fear. Break down the fear into smaller components and expose student to them slowly. **Displacement** Directing emotions on something different than the causer **Rationalization** Find excuses for poor performance. Student fully believes excuses. **Fantasy** Fantasizing about success rather than taking measures to achieve it (daydreaming). Affects motivations **Compensation (lifted truck theory)** Counterbalance a weakness with success in another area **Projection** Projecting a blame. ATC was talking while I was landing which messed up my landing. Instructors might project blame on students when they failed to teach. **Reaction Forming** Forming fake attitudes to counter internal desires. Believe the opposite is true.
Learner emotional reactions	MANY stress and anxiety inducing situations. There are a lot of things to promote anxiety (solo, solo xc, checkride…) Student, you will have anxiety about these things! Instructors cannot cure anxiety

Anxiety and Stress	**Anxiety** Anxiety is a feeling of worry, nervousness, or unease, often about something that is going to happen, typically something with an uncertain outcome. Can lead to impulsive thinking or lack of action "You have made effort and done so well you will do great" **Stress** Normal and abnormal reactions Normal stress is healthy and can promote focus. **Abnormal**: **PRESIM** - Painstaking self control (death gripping yoke) - Rapid change in emotions - Extreme over cooperation (ATC says 20 left into TS) - Severe anger (anger at switching planes) - Inappropriate laughter or singing - Marked changes in mood on a particular lessons (angry when assigned NAVLOG) Don't mess up my pass rate…Don't break the airplane…not good ways to handle anxiety. There are good thing and bad things with stress
Impatience	- The impatient learner fails to understand the need for preliminary training and seeks only the ultimate objective without considering the means necessary to reach it. - The instructor can correct learner impatience by presenting the necessary preliminary training one step at a time, with clearly stated goals for each step. The procedures and elements mastered in each step should be clearly identified when explaining or demonstrating the performance of the subsequent step. - Impatience can result from instruction keyed to the pace of a slow learner when it is applied to a motivated, fast learner. ADVANCE learners to the subsequent step as soon as one goal has been attained
Worry or Lack of Interest	Similarly, worry or lack of interest also hinders the learning process. Instruction should be keyed to the utilization of the interests and enthusiasm learners bring with them, and to diverting their attention from their worries and troubles to learning the tasks at hand.
Physical Discomfort, Illness, Fatigue, and Dehydration	THINK MASLOW'S hierarchy Specific dangers with **Fatigue -** most treacherous hazard to flight safety since you may not notice it until serious errors are made. **Acute or Chronic fatigue** Acute fatigue is characterized by: • Inattention • Distractibility • Errors in timing • Neglect of secondary tasks • Loss of accuracy and control • Lack of awareness of error accumulation • Irritability **Chronic fatigue** occurs from repeated episodes of acute fatigue. Recovery takes a prolonged and deliberate approach.
Apathy due to Inadequate Instruction	Instruct at the level of the student.
Teaching the adult learner	- Help learners integrate new ideas with what they already know to ensure they keep and use the new information. - Recognize the learner's need to control pace and start/stop time.
Effective communication	Communication is the key for me as an instructor.
Basic Elements of Communication	**SSR** (source, symbol, receiver) Basic psychology dictated three basic elements of communication. If one element is ineffective, communication fails The SOURCE must understand the RECEIVERS **abilities**, **attitudes**, and **experiences**.

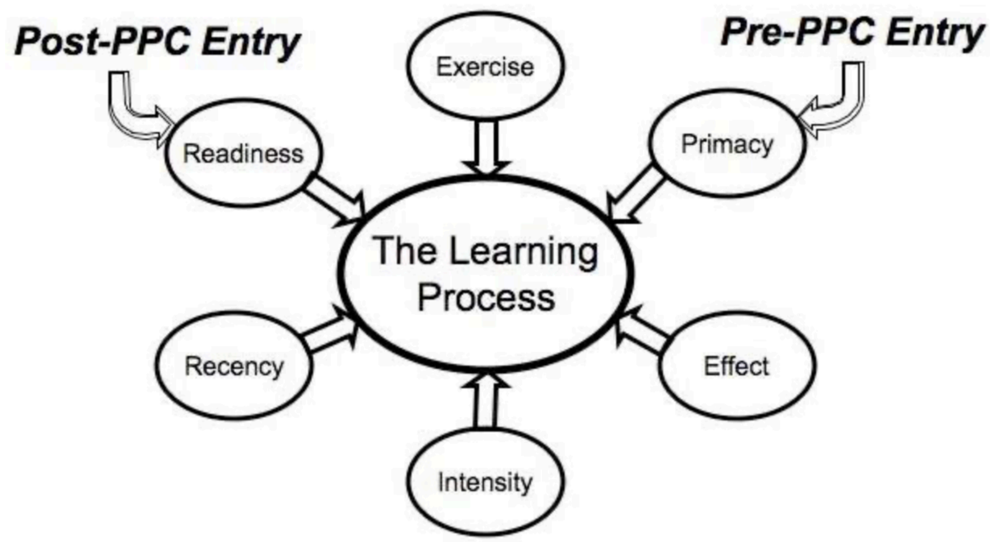
Barriers to Effective Communication	**COILE** (confusion, overuse, interference, lack of common experience) But along the way, there are barriers that bounce back my attempts to communicate. **Confusion between symbol and symbolized object** Do not allow any room for misinterpretation. Instead of using your hand as an airplane, use an airplane model. **Overuse of abstractions** Make sure to be specific and concrete in explaining topics. Rather than "more RPM" or 'hold the nose up a little', say "increase RPM to 2300" and then relate it to the big picture afterward. **Interference** Noise can disrupt message making the learner hear something different. Landscapers outside, poor headset microphone… **Lack of Common Experience** Single greatest barrier. Words are not like bricks being delivered on a truck. Rather, they stimulate the learner to make a connection to previous experiences and form new ones. Having a disconnect between what the instructor wants to convey and what the student takes away is a massive hinderance to learning. Ex. Using port and starboard but they dont have boat experience **External Pressures** Maslow!
Developing Communication Skills	But how do we get better at communicating: **LIQIR** **Listening** Hearing with comprehension and watching their physical behavior. **Instructional Communication** Have coherent and organized thoughts **Questioning** Instructors should ask focused, open ended questions (why, how) and avoid closed-ended questions. Closed ended questions focus on rote knowledge. **Instructional Enhancement** A good instructor is always learning, and thus getting better **Role Playing** We will see this in the runway incursion lesson

Today we talked about many things an instructor can do to better work with their student in regards to knowing their needs and motivations. If an instructor can do this, they will grow and can better serve the needs of their students.

LET ME MAKE SURE I COMPELTED ALL ITEMS. DO YOU HAVE ANY QUESTION

B. THE LEARNING PROCESS

Definitions of learning.	**A change in behavior as a result of experience.** If you dont have experience, there is no way you could've learned.
Learning theory as it applies to instruction	We have these theories on how people learn.
Behaviorism	Explains human behavior as cause (stimuli) and effect (response). Best example is Pavlov's dogs. They were conditioned to associate bell with eating (and salivating). But this also applies to people: with positive and negative stimuli.
Cognitive Theory	**Cognitive: related to thought; "why did they do that?"** Humans best learn when relating new knowledge to existing knowledge. Best to REFLECT
Other	**Information** Processing - computer. **Constructivism** - building upon previous knowledge.
Perceptions and insight.	Senses lead to perceptions, grouping perceptions lead to insights Now that we've discussed how we process stimuli and experience around us, lets talk about perceptions and insight. Perceptions are how we sense the world around us. It is important to properly teach students associations (smell of AVGAS to identify leak, meaning of annunciator light), they NEED those perceptions to make connections. We must tell student everything that goes on. Improper perceptions lead to errors. The more senses, the better the experience. However, it isn't so clear. There are factors that influence perception include: **GSTEP** (RELATED TO MASLOW) **Goals and Values** - Students values directly color how they interpret events. They may seek info that supports own belief **Self-Concept** - related to self-esteem. "All is lost" negative self image may skew perceptions. "Spins are unsafe and I just will stay coordinated" may see the lesson as negative. Learners who view themselves positively, on the other hand, are less defensive and more receptive to new experiences. **Time and Opportunity** - Adequate time and experience is imperative to form accurate perceptions. You cant say read the airplane flying handbook and land the plane tomorrow. Student has no experience yet! **Element of Threat** - Feeling unsafe can trigger fight or flight. **Physical Organism** - Five senses are malfunctioning (sick, forgot glasses) can't properly perceive. When we have multiple perceptions, we can form insights. **Insights are perceptions grouped together to form a meaningful whole.** This is an **instructors responsibility.** Related hold short line (visual perception) and runway incursion avoidance (insight and building blocks).
Acquiring knowledge.	So far, we've discussed the theory of learning, but is there a specific step to do to actually learn things? YES. **Steps to acquire knowledge (MUC)** **1. Memorization** - this is where we start (ABCDE…), no meaningful insights but its a start to knowing something (MUST STOP AT HOLD SHORT) **2. Understanding** - Organize memorized thoughts to form mental models (Why do we stop at hold short) **3. Concept learning** - use what you know to solve problems and make decisions. (ADM)

Laws of learning.	If we make use of the following laws of learning, it will help out students learn more effectiely. **REEPIR** **Readiness** - NOT ready for lesson with pen and paper, but MENTALLY ready with clear objectives. If not, then lesson is useless. **Excercise** - use lesson often, USE IT OR LOSE IT **Effect** - learning experience in enhanced with positive experiences, and weakened with negative experiences. **Primacy** - things learned FIRST are best remembered. **Intensity** - real world applications are better at enforcing lessons. Make lessons vivid. **Recency** - How long did you learn something 
Domains of learning, including:	CATEGORIES in the cognitive theory.
Cognitive **Book smart pilots**	What someone is thinking and how they acquire knowledge. As related to the steps of acquiring knowledge, four LEVELS of learning **RUAC** **Rote –** Memorizing a fact. P factor makes the airplane turn left **Understanding –** Understand why something occurs. In a climb, propellor on the down-moving blade moves faster so we have a turning tendency. **Application –** Putting something to use that has been understood. This causes us to need right rudder (ACTUALLY DO IT) but we are not done.. !**Correlation –** Applying what was learned with subsequent or previous learning. Applying slow flight techniques to landings. THIS WILL HELP YOU IN THE FUTURE BECAUSE… REQUIRES ANOTHER LESSON TO RELATE TO
Affective	Deals with how a person feels about their training as they go through it. Imagine a new student. These are the stages they progress through. **ARVOI** **Awareness** – they are open to listen and begin learning (law of readiness high) **Response** – person begins to interact and **participate**. Asking questions **Value** – student sees the value in the training **Organizing** – student organizes training into personal belief system. They believe that if they use the lesson it will help them. **Integration** – Not only is the student internalized what they learned, but they integrate the lesson into their life.

Psychomotor **Stick and rudder pilots**	Those cover what goes on in their brain regarding thoughts and emotions, but what about the physical execution? Psychomotor is a **skill** based domain. Development of these skills utilizes repetitive practice and is measured in terms of speed, precision, distance, and techniques. **OIPH** **Observation** – Student sees instructor demonstrate maneuver within limits **Imitation** – Student tries it themselves **Practice** – with or without instructor **Habit** – Twice the time as a professional
Characteristics of learning.	We will notice there are certain characteristics of learning. These are more abstract than learning itself (**PEMA**) **Purposeful** – you must have some kind of purpose to learn. **Result of Experience** – a change in behavior as a result of experience **Multifaceted** – involving multiple senses, the more the better. **Active Process** – you must be mentally ready and willing to devote the time and participate
Scenario-based training (SBT).	Have a clear set of objectives. Plan XC to airport with too high elevation. SBT has a clear set of objectives, is tailed to the needs of the learner, and capitalizes on the nuances of the local environment
Acquiring skill knowledge:	Knowledge that manifests in the ability to do something (different than knowing theory)
Stages	**CAA** **Cognitive Stage** – Having the factual knowledge to know how to conduct the task. "What am I supposed to do next?" **Associative Stage** – Student can assess their own performance, although performance may not be perfect We really want them at the final stage **Automatic Response –** Student can perform without any input and it requires less attention.
Knowledge of results	The most important thing to remember is it is easier to correct a mistake than to unlearn something incorrect.
How to develop skills	**LAW OF EXERCISE and POWER LAW OF PRACTICE** In first trails, coordination is lacking. Mistakes are frequent, but each trial presents things to improve.
Learning plateaus	- Learning tends to progress quickly in the beginning, but then may reach a plateau. Move the learner to a different place in the curriculum. - They could also be a period of repetitive improvement/decline **OVER PRACTICE / CAPABILITY LIMIT / LACK OF INTEREST** - **change it up, but get them excited again (cross country for lunch, etc)**
Types of practice.	**DBR** **Deliberate Practice** – Specific practice on a particular portion of a lesson. Maybe practicing roll in and roll out of steep turn **Blocked** – Repeat lesson but with no idea of concept of lesson. (Student becomes a robot on dealing with task) **Random** – Having a variety of practice. For example, diversions don't happen in the same place. Aborted takeoffs happen for many reasons.
Evaluation versus critique.	In the beginning stages of skill acquisition, it is more helpful to provide practical suggestions rather than an evaluation or grade. When the learner is later performing to meet standards, evaluation becomes more useful.
Distractions, interruptions, fixation, and inattention.	Appropriately timed distractions can help students develop the sense for what things can be deferred and what requires immediate attention.

Errors	Along the way, people will make mistakes and have errors. Two types **Slips –** Plans to do one thing, but inadvertently does another. Error of action **Mistakes** – Plans to do something and is successful, but action was itself incorrect. i.e. ignoring traffic calls in pattern since its usually busy Minimizing Errors (**DR CULT**) **Develop Routines and standardized flows** **Raising Awareness –** We should be more situationally aware of our surroundings during error-prone operations (fatigue, out of practice) **Check for Errors** – Be alert to what is going on **Using Reminders** – Heading bug, fuel timers **Learning and Practicing** – higher knowledge and skill will lead to fewer mistakes **Taking Time and don't Rush** – Never trade speed for accuracy
Memory, including:	Speaking of errors, lets discuss memory and forgetting things. We have three types memory.
Sensory	See it, touch it, smell it, it comes and goes pretty quick (initial stimuli). Sensory memory is the initial filter for taking in an event, filtering out extraneous info, and passing it to the short term memory.
Short and Long-Term Memory	Information is stored here for ~30 seconds, after which is is lost or transferred into long term memory. Goal of short term memory is to put information to immediate use. A better way to retain information is to categorize or chunk it into more manageable sections. Iconic memory (remembering pictures), acoustic memory (sounds), and working memory, which functions like a scratch pad Permanent storage of unlimited information. If significant is attached to a memory in the sensory or short term phase, it becomes much easy to recall in long-term.
How usage affects memory	**Frequency and recency –** will be retrieved easily and quickly. Ideal situation. **Frequency only –** hard to retrieve or not retrieval at all. Some recent rehearsal is needed to refresh **Recency only –** vulnerable to being thrown away for lack of significance. Requires a program of regular rehearsal. Cramming is ineffective for long-term.
Forgetting	**Retrieval Failure** – unable to retrieve information stored **Interference –** Two memories are overlapping (moving from one plane to another) (one pedal driving to regular) **Fading** – Information used for a while can decay **Repression or Suppression** – A memory is pushed away
Retention of learning.	So what can we do to remember things better? **MR LAMP** **Meaningful repetition aids recall** – **Recall prompted by association** – **Learn with all senses** – See multifaceted **Attitudes that are favorable** – **Mnemonics** – Acronyms (ATOMATOFLAMES) Acrostics () Rhymes () Chaining () **Praise will stimulate recall** – Positive reinforcement!
Transfer of learning	**Positive Transfer** – Prior knowledge helps completion of new task **Negative Transfer** – Prior knowledge hinders completion of new task (driving to taxiing) **Near Transfer** – Applying knowledge to a closely related lesson **Far Transfer** – Applying knowledge to a distant lessons **We must plan for this!**

As instructors, we must capitalize on these theories behind learning to ensure our students have the best chance at receiving, retaining, and using new knowledge.

LET ME MAKE SURE I COMPELTED ALL ITEMS. DO YOU HAVE ANY QUESTION

C. THE TEACHING PROCESS

Teaching	To instruct or train
Process	Preparation, presentation, application, and assessment. We will cover these in our discussion today.
Essential skills	**PAMS** **People Skills** – The instructor student relationship must be positive. **Assessment Skills** – not only asking questions, but assessing student commitment to flight training and ability to continue training **Management Skills** – Managing yourself, student, time, and expectations. Imperative to be an effective instructor **Subject Matter Expertise** – You know what youre talking about
Course of training.	Complete series of materials to achieve a goal, usually a rating. Each lesson requires **OBJECTIVES** and **COMPLETION STANDARDS.** Curriculum – multiple rating outline (think 141 degree) Course – Outline for one rating Syllabus – Outline for what lesson is prepared when. Maybe a ground and flight portion
Preparation of a lesson, including:	Lesson plan is the outline required for a particular lesson
Training objectives and completion standards	It is important for the student to understand the criteria required to continue to the next lesson
Performance-based objectives	Have specific **description**, **condition**, and **criteria**. "Using appropriate charts, a navlog, and training aircraft, navigate from point A to B while maintaining ACS altitude tolerances and 5 minutes ETA."
Importance of Airman Certification Standards (ACS) in aviation training curricula	
Decision-based objectives	Allow more scenario-based and dynamic objectives. Promotes ADM. "Student can pick an appropriate landing site under conditions of simulated engine failure."
Organization of material.	For the lesson itself (not necessarily the plan) **INTRODUCTION (AMO)** **Attention –** Have you ever used a GPS in your car. Well we have them in plans, but they are much more sophisticated. **Motivation** – If you know how to properly use this, you'll never get lost! **Overview** – Today we will talk about how to turn it on, basic layout… **DEVELOPMENT** **Past to Present** – if history is important in the lesson, arrange the lesson chronologically **Simple to Complex** – Organized by complexity, start with simple (Airspace) **Known to Unknown** – Start with something known by the student (traffic pattern to rectangular course) **Most used to least used** – for navaids, start with GPS, then VOR, then NDB **CONCLUSION – then we have to wrap it all up** *also check lesson and ask for questions* Tell them what youre gonna tell them, tell them, tell them what you told them.
Training delivery methods, including:	**LGECCDD**

Lecture	Introduce new subjects, summarize ideas, show relationship between theory and practice, and reemphasize main points. **IBFIT** **Illustrated talk** – Use training aid to walk through discussion **Briefing** – Clear and concise direct facts **Formal Lecture** – Little interaction with students **Informal Lecture** – in a somewhat formal environment, but there is some participation **Teaching Lecture** – Much interaction with hands on and projects Benefits – many students at once (most economical). Help supplement other methods. Many ideas at once Disadvantages – not effective for learning large amounts in short time. No way of checking student process **PREPARATION** Establish objectives and desired outcome Research subject Organize material (lesson plan) Plan productive classroom activities Suitable language (define technical terms) **TYPES OF DELIVERY** Manuscripts Memory Extemporaneously from outline (BEST) Impromptu without preparation
Discussion	
Guided discussion	Uses a series of questions to promote involvement What kind of questions **(HCCS RR)** **Have a specific purpose** **Clear in meaning** **Constant single idea** **Stimulate thought** **Require definite answer** **Relates to previously covered information**
Cooperative or group learning	People can work together to maximize each others learning capacity Those who participate have higher test scores, higher self esteem, improved social skills, and greater comprehension of the subject
Demonstration-performance	Most common form IN AIRPLANE. Here, **EDSIE** **Explain** **Demonstrate** **Student performs** **Instructor supervises** **Evaluation** Most widely used teaching tool in airplane and used often in ground Students performance is only as good as instructors teaching
Drill and practice	Should be done every time. Repetitive questioning to enforce Promotes learning through repetition. LAW OF EXCERCISE

Electronic learning (e-Learning).	Very time flexible Cost competitive Learner centered Easily updatable Accessible anywhere and anytime Lack of pure interaction Difficult for instructor to ration Difficult to find good programs for subject material Expense of equipment Lack of technical knowledge
Instructional aids and training technologies, including:	**Support** and **reinforce** what has been taught. Does not replace lesson.
Characteristics of effective instructional aids	Aids should be simple and compatible with the learning outcomes to be achieved. Obviously, an explanation of elaborate equipment may require detailed schematics or mock-ups, but less complex equipment may lend itself to only basic shapes or figures.
Reasons and Guidelines for use	Instructional aids help gain and retain learner's attention and clarify the relationships between material objects and concepts. The instructional aid should be selected based on its ability to support a specific point in a lesson. When an instructor finds themself forming visual images with words, instructional aids should be considered.
Types	Includes boards, models, videos…
Integrated flight instruction.	
Problem-based instruction.	Promote ADM with SBT. Students reach real world solutions to real world problems. **SBT (SDWINDRS)** Should promote situational awareness and opportunities for decision making Does not have an obvious answer Will not have one right answer Is not a test (no one answer) iNteractive Domains (all 3) Require time-pressured decisions Should not promote errors **Collaborative Problem-Solving Method** **Case Study**
Planning instructional activity, including:	
Blocks of learning	It is imperative to identify the specific elements required for a training course. For example, blocks may include ground ops as a fundamental component of training. This helps break down intimidating goals into attainable steps.
Training syllabus	Blocks of learning to be completed in the most efficient order. As a new CFI, it would be best to purchase these.

Lesson plans	**OEEISC** **Objective** **Elements / Content** *schedule* **Equipment** – **IP Actions** – Detailed, “the instructor shall demonstrate the use of the E-6B by showing its features…” What specifically are you going to do. **SP Actions** – Student shall participate in the calculations of the E-6B to ensure they understand its use **Completion Standards** – Preflight and perform remedial tasks on Garmin GPS 175 which are essential for VFR operations. Characteristics of a well planned lesson - unified segment of instruction (use as many resources and instructional styles as possible) Maximum insight potential - Content - each lesson should contain new material - Scope (should be narrow to focus on desires lesson) - Practical – Planned considering how the lesson will be delivered (in air, ground…) - Flexible – may need to deviate depending on student - Relate to course of training – have risk management sections - PPAR (Prepare, present, application, review (through assessment) SBT should be injected into every lesson!

D. STUDENT EVALUATION, ASSESSMENT, AND TESTING

Purpose and characteristics of effective assessment.	**Assessment -** Process of gathering measurable information to meet evaluation needs **Purposes** **- Identify student deficiencies** **- Provides feedback to client (they can see their own performance)** **- Help students evaluate their own performance (How do you think you did… self critique)** **- See where more emphasis is needed** **Caracteristics** **FASTCOCO** **FLEXIBLE** – fit tone, technique, and content to occasion. There is no formula. Cant be too hard in the beginning of training, can't be to easy toward the end of training. **ACCEPTABLE** – Critique must be accepted. If the student does not accept where the critique is coming from, they won't accept the critique. **SPECIFIC** – must be specific in what needs fixing. "Your second landing wasn't as good as the first" is not helpful **THOUGHTFUL** – Maslow's hierarchy and need for self esteem. Respect the feelings of the student, and maybe don't deliver in front of other people… **COMPREHENSIVE** – cover weaknesses and strengths. **OBJECTIVE** – no personal opinions, base critique on fact **CONSTRUCTIVE** – give positive guidance for correction. **ORGANIZED** – Assessment should make sense. Don't go out of order.
Traditional assessments.	There are a couple ways we can assess. We have all experienced the first type, a traditional assessment. This is a written test. **Selection Type Questions** True false, ABCD, making a selection **Supply Type Questions** Very subjective. Short answer / essay style. Two instructors may grade differently. Pre-solo written is usually the first assessment someone gets. Making your own **DRCOVU** **DISCRIMINATION –** degree to which test distinguishes the difference between students (We want it discriminate…that way we can guide where people are at) **RELIABLE** – degree to which test results are consistent with repeated measurements, see ATC giving different directions **COMPREHENSIVE** – degree to which test measures overall objectives. In pre-solo, we want to cover respective regs in 61 and 91, airspace student solo in, and systems student will use. We don't want to go overboard, however. **OBJECTIVITY** – essay style questions are less objective than selection type **VALIDITY** – degree to which test measures what it is supposed to measure, and is the most important. Give test to peers to review. **USABILITY** – refers to functionality of tests. Fonts, graphics, etc.
Authentic assessments, including:	**STAGE CHECK** Asks the student to perform real-world tasks and demonstrate a meaningful application of skills and competencies. SBT!
Learner-centered assessment	**REPLAY –** walk me through the lesson. Helpful because you can identify between instructor way of seeing things and student way of seeing things. Resolves discrepancies. **RECONSTRUCT** – ask student to identify what could be seen done differently. **REFLECT –** Ask why/how. Where did you go wrong? PLACE MEANING ON EXPERIENCE **REDIRECT** – Form insights by relating to other experiences, and how that might help future sessions.

Maneuver or procedure grades	Rubric **DESCRIBE** – Learner can explain the desired performance **EXPLAIN –** Learner can explain the underlying concepts and principles **PRACTICE** – Learner can perform with assistance **PERFORM** – Learner can perform without assistance
Assessing risk management skills	Rubric **EXPLAIN** – Learner can explain the risks and hazards **PRACTICE** – Learner can implement SRM controls into performance with assistance from instructor **MANAGE-DECIDE** – Learner can identify and perform adequate ADM
Choosing an effective assessment method.	1. Determine level-of-learning objectives. 2. List indicators of desired behaviors. 3. Establish criterion objectives. 4. Develop criterion-referenced test items.
Purposes and types of critiques.	**ISSISW** INSTRUCTOR/STUDENT CRITIQUE – members of class are invited to critique someones performance. Instructor leads discussion STUDENT LED CRITIQUE – students lead assessment instead of instructor. Not efficient. Students may be interested however. SMALL GROUP CRITIQUE – class is divided into small groups, each analyzing a specific skill or portion of the lesson a student gives. One person from each group presents their finding. Leads to a well rounded discussion since there are more diverse responses. INDIVIDUAL STUDENT CRITIQUE BY ANOTHER LEARNER – one person presents entire critique SELF CRITIQUE – most common. “How do you think you did”. NOT like collaborative assessment/4 r’s. WRITTEN CRITIQUE – Good. Instructor can devote time to critique. Students can hold on to comments. Student has many comments from a lot of people. Drawback is that no one else benefits since there is no collaborative environment. **AVOID** Don't be too long. Avoid absolute statements (landing was bad). Avoid controversies. Don’t get in a situation where you have to defend your critique.
Oral assessment, including:	Simply asking student questions. Reveals effectiveness of instructors instruction. Check students retention. Good for reviewing previous lessons from previous times. Emphasize important points of training. Promotes active student participation.
Characteristics of effective questions	**ABCAP** **Apply** to the subject of instruction **Brief** and concise, but clear and definite **Centered** on one idea with one correct answer **Adapted** to the ability, experience, and progress of the student (don’t ask about the thickness of the p-lead of a magneto) **Present** a challenge to the student (takes a while to get to know student, however) engineer type student vs non-engineer type

Types of questions to avoid	**POTBIT** **Puzzle:** "What is the first action you should take if a conventional gear airplane with a weak right brake is swerving left in a right crosswind during a full flap, power-on wheel landing? **Oversize: "**what do you do before beginning a flight" **Toss-up** "In an emergency, should you squawk 7700 or pick a landing spot?" YES **Bewilderment** "In reading the altimeter—you know you set a sensitive altimeter for the nearest station pressure—if you take temperature into account, as when flying from a cold air mass through a warm front, what precaution should you take when in a mountainous area?" What now **Irrelevant:** asking irrelevant questions (see appendix b) **Trick questions:** A,B,C,D are reversed, or not questions
Answering learner questions	1. Make sure you understand clearly 2. Display interest. You may hear the same questions over and over again… Answer directly and accurately as possible 3. Determine satisfaction with answer. Does that answer your questions? If a questions is too advanced, don't be afraid to push it until a more relevant time. Also admit when you don't know an answer to a question
Assessment of piloting ability.	

E. ELEMENTS OF EFFECTIVE TEACHING

Aviation instructor responsibilities, including:	**HEMPSE**
Helping learners	Help students learn and should be enjoyable. This helps maintain a high level of motivation. More motivation yields a more enjoyable experience. We can do this by clearly defining standards they can work to. Setting standards allow students to feel capable of doing something
Emphasizing the positive	Instructors conduct greatly influence students behavior. Lead by example and stay away from negatives. Fear has a significant effect on perceptions, and positive instruction results in positive learning.
Minimizing learner frustrations	**MACKBAG** Most pilots want to be pilots and not students, so being able to see progress and an end in sight is imperative. **M**otivate learners – Solo celebrations, emphasize benefits. **A**pproach learners as individuals – No two students learn alike **C**riticize constructively – Critique without explanation is frustrating **K**eep learners informed – Learners feel insecure when they don't know what is expected of them. Also give them overview of course. **Be** consistent – This only leads to student confusion **A**dmit error – Learners can sense when an instructor doesn't know what they are talking about. **G**ive credit when credit is due – Praise provides incentive to do better and lets student know of their progress
Providing adequate instruction	Adequate referring to tailor lesson to learner. **Fast Learners** - present unique difficulties: they make few mistakes and thus ignore the importance of correcting errors. Instructor should raise standards of performance but not give unfair criticism for the sake of it. **Slow Learners** - assign more easily attained goals. Before attempting a complex task, the instructor separates it into discrete elements, and the learner practices and becomes good at each element
Standards of performance	These let students know exactly is expected. However, you wouldn't expect a lesson 1 student to hold ACS standards of performance. Standards for taxi may be "keep it on the pavement" or "keep it in the center of the sky." Hone in standards as skill progresses. Standards are just an accessing tool, not a teaching tool.
Ensuring aviation safety	Students will mimic instructor. Safe practices will get replicated by student.
Flight instructor responsibilities, including supervision and surveillance during training.	**PEEPPASE** **P**ilot supervision – FI has a responsibility to supervise pilot. **E**valuation of SP ability – use all tools in assessment and critique. Compare ability to standards or common sense (you don't have to say anything) **E**ndorsements – **P**ractical test recommendation – IACRA **P**hysiological obstacles – Vibrations, g-forces, noise, are probably unfamiliar to students. Don't ridicule them for nervousness. (Sickness, end lesson asap since they wont learn well - Maslow's hierarchy) **A**dditional training and endorsements – flight reviews, pilot proficiency **S**ee and avoid responsibility – emphasize LOOKING OUTSIDE to students and help them during initial training. **S**olo thought process – **E**nsure student skillset –As required by part 61 and ACS for particular objective (pre solo, checkride…)

Flight instructor qualifications and professionalism.	**SADPP** **S**incerity – honesty, straightforward admit mistakes, find answers to their questions **A**cceptance of student – Every learner is different, fast/slow…accept their learning style. **D**emeanor – positive attitude, good demeanor **P**ersonal appearance and habits – neat, clean, and appropriately dressed. You are being PAID. Look professional **P**roper language – No profanity. Speak positively and descriptively
Professional development.	Instructors should maintain a high level of expertise in their field. WINGS, NAFI, Gold Seal, AOPA ASI, type clubs (Cirrus owners…, Facebook group, participate in noticed for proposed rule making (NPRM) … WINGS is the only approved continuous training program. FSINS. 8900-1
Instructor ethics and conduct.	https://www.secureav.com/FIMCC-v1.0.pdf

F. RISK MANAGEMENT

Teaching risk identification, assessment, and mitigation.	Risk management is a decision-making process designed to identify hazards systematically, assess the degree of risk, and determine the best course of action. Goal is to proactively identify safety related hazards and mitigate potential risks. **Severity vs probability (day vfr vs night imc)** - probable occasional remote improbable - Catastrophic critical marginal negligible **Hazard –** a present condition, event, or circumstance that could lead or contribute to an undesirable event. A source of danger. Like a crack in the airframe **Risk – The future impact of a hazard that is not controlled** ADM are the decisions we make for right now, risk management is the decisions we make for the future. **Principles of Risk Management (AMAI)** **Accept** no unnecessary risk (I.e. accept only necessary risk) **Make** risk decisions at appropriate level (in single pilot ops, ATC should not be calling the shots. You're responsible for 5 p's) **Accept** risk when benefits outweigh cost (balance scale, new type but good weather) **Integrate** risk management into planning (think about risks early) **Risk management Process** (IAM) 1. Identify the hazard (i.e. deteriorating weather already at personal minimums) 2. Assess the risk. Assess considering likelihood and probability. 3. Mitigate the risk. Investigate strategies that reduce, mitigate, or eliminate the risk. (Check weather along route, plan alternate, file IFR)
Teaching risk management tools, including:	These checklists are supposed to become second nature.
PAVE (proactive)	**Pilot -** Currency and IMSAFE **Aircraft** – **Equipped** for trip (AVIATES, GRABCARD, ATOMATO…) **Environment** – Weather, airspace, airports, day/night, terrain **External pressures** – Passengers
IMSAFE	Good for physical and mental readiness
PPP (reactive)	Perceive – see annunciator light Process – What action do I need to take Perform – do mitigating action
5 P's	5 variables that should be consulted at key decision making moments Plan - the mission, cross country planning, and weather Plane - is the plane capable for whats being asked of it Pilot - IMSAFE Passengers - discuss options and clearly outline risks. Don't fall victim to self-induced pressures Programming - stay ahead of the airplane
TEAM	Transfer, eliminate, accept, and mitigate.
DECIDE	
Personal minimums checklist	Gets student thinking about what to consider since they dont know whats a risk
Flight Risk Assessment Tools (**FRATs**)	A fixed list of hazards and associated risks are presented and assigned "scores" based on the severity of the hazard. Typically, if the total score is below a certain number, the pilot can begin the flight. If the score is above a certain number then some sort of mitigating action is required. DANGERS- singular items may pose an extreme hazard and are not captured (line of TS)
When and how to introduce risk management.	Risk management should be discussed at the beginning of training!

Risk management teaching techniques by phase of instruction.	**Presolo -** Instructor-led and guided risk management training should occur during pre-solo instruction. Risk management should be part of every preflight and postflight brief. **Post Solo prior to Cross Country Training** - During the initial solo and dual flights following the first solo, the learner should be able to perform a risk analysis of the planned flight, with occasional coaching from the instructor. **Cross Country Training -** During the cross-country phase of training, learners should master risk management techniques commensurate with the complexity of flights and the terrain along the route to the destination(s). This could include, for a private certificate, flights at night, flights in complex airspace, or to unfamiliar airports. Instructors should ask the learner to accomplish a full risk analysis for every dual and solo cross-country flight. The process should include use of a FRAT or other method of analysis
Managing risk during flight instruction, including:	**ABCSP** - Ask the learner to fly specific maneuvers after giving appropriate training. - Be prepared to take over the control of the aircraft. - Choose practice locations that provide safe options. - Stay alert for the unexpected either from the learner or external elements. - Perform maneuvers with sufficient altitude.
Common flight instruction risks	**P** - Currency: any unfamiliar in equipment creates risk. IMSAFE: must be aware of self and SP **A -** Flight school aircraft may be abused, and not always flown by same instructor **V -** Practice areas may be crowded. Stalls to full break increase risk of spins **E -** Stress and anxiety of student
Best practices	**P** - Currency/Proficiency: read POH. IMSAFE: must be aware of self and SP **A -** IP/SP should review airworthiness and discrepancy log. Learn how to speak to maintenance **V -** Discuss with student the environment **E -** Ease student's concerns if aviation related
Special considerations while teaching takeoffs and landings	- Do NOT try to teach the student while in takeoff or landing, they are already task saturated. - Teach specific hazards - wake turbulence, separation, and crosswinds. - Incorporate scenario based training
Aeronautical Decision-Making (ADM) to include using Crew Resource Management (CRM) or Single- Pilot Resource Management (SRM), as appropriate.	Art and science of managing all the resources available to a single pilot to ensure the successful outcome of the flight. Accommodate but don't sacrifice personal minimums for passengers.
Operational pitfalls	Bad habits arising from misplaced priorities (get-there-itis, scud running, inadequate fuel reserves)